

Center for AI in Science and Engineering (ARTISAN)

Integrating People, Software, and Hardware to
accelerate AI-Driven Discovery

Suresh Marru

Director, ARTISAN

Research Professor, College of Computing



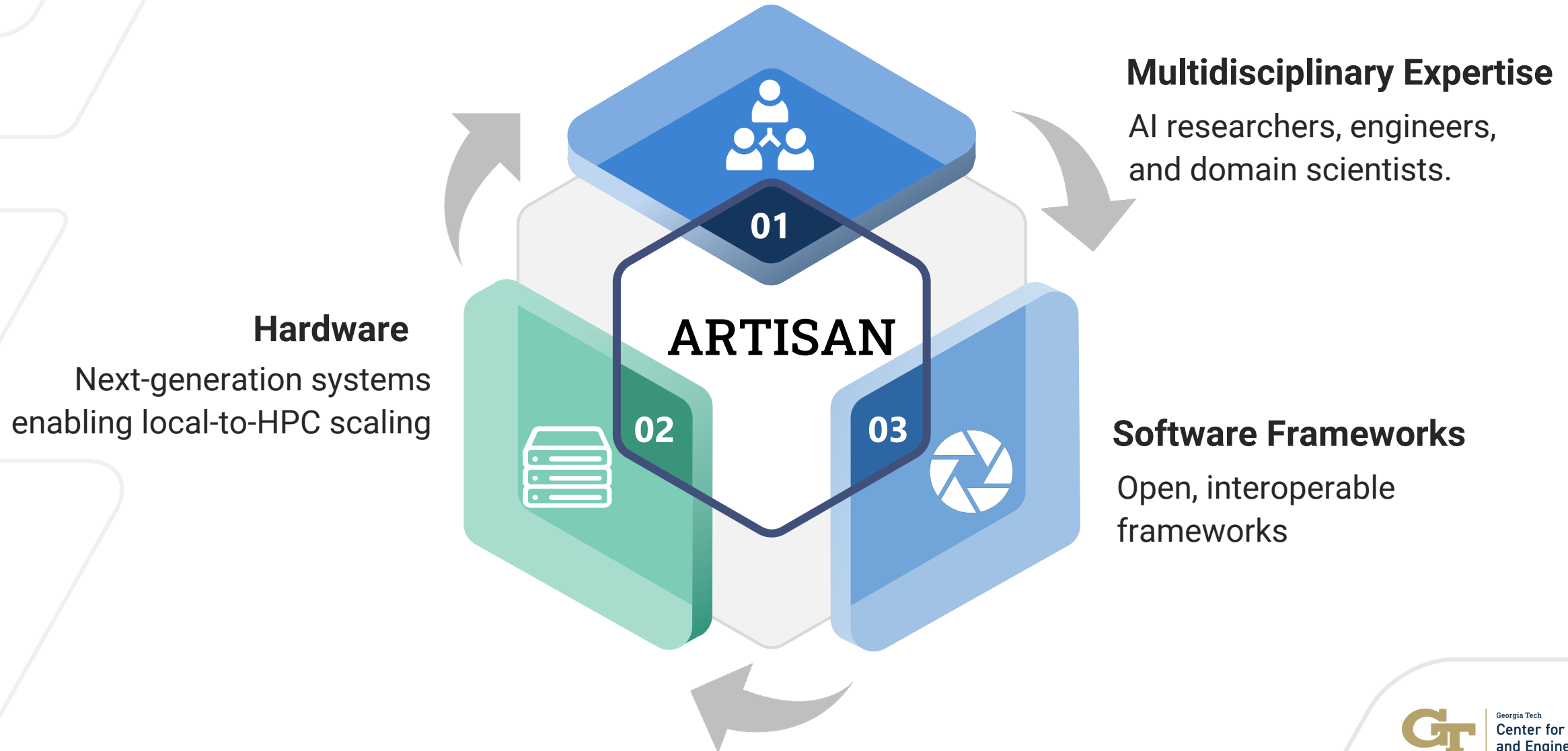
Georgia Tech
Center for AI in Science
and Engineering

**A joint Center between the
College of Computing (CoC) and
the Institute for Data
Engineering and Science (IDEaS)**

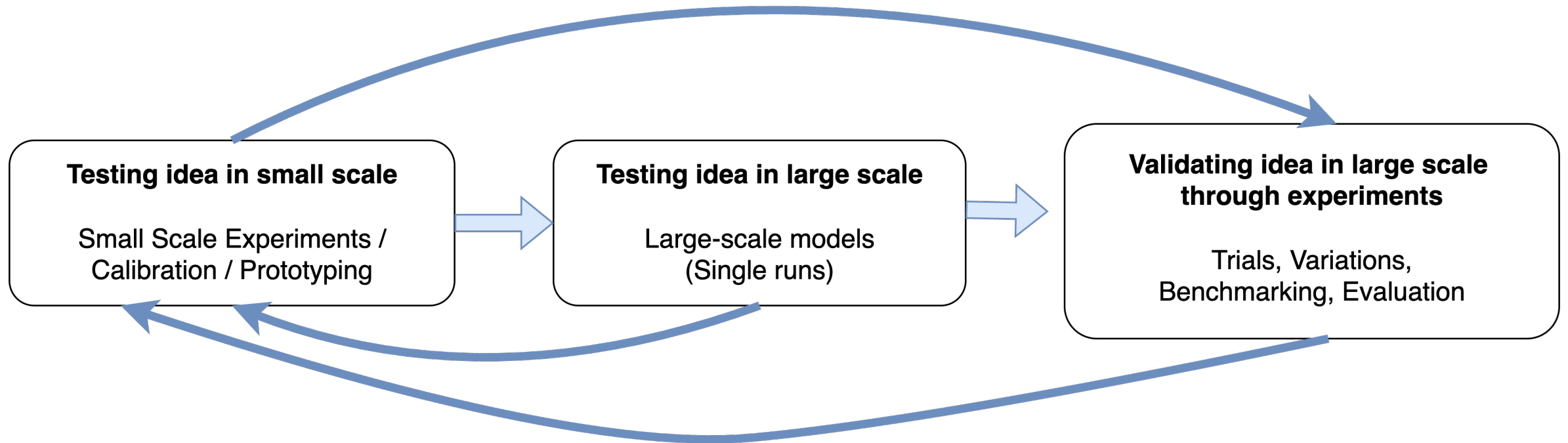


Georgia Tech
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ARTISAN: An Integrated ecosystem for AI-driven Innovation



Multi-Scale Computational Science



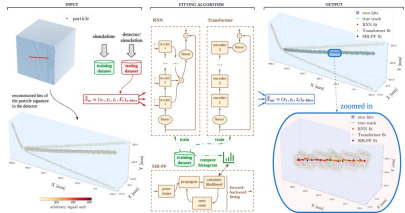
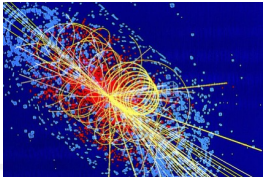
Modern science spans from laptop-scale exploration to exascale simulation, but software and hardware gaps limit seamless scaling

Scaling Bottlenecks Limit Progress

Scientific Domain	Typical Scaling Challenge
Molecular Dynamics	Transition from short equilibrium runs to large production runs; Multiple replicas; Enhanced sampling methods
Quantum Mechanics / Density Functional Theory	Screen large molecule sets from initial small molecule studies; Evaluate multiple computational methods
Neuroscience	Large-scale parameter sweeps; Multiple trial replications; Scaling from single-cell to full network models
AI for Science & Engineering	Train/test from small to large architectures; Scale up datasets for engineering simulations
Deep learning / AI	Transition from small to very large models; Expand from few to many datasets

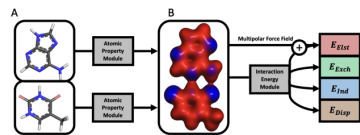
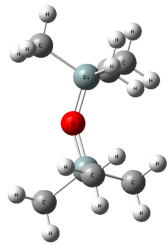
Large (Foundation) AI models for Science & Engineering

Particle Physics



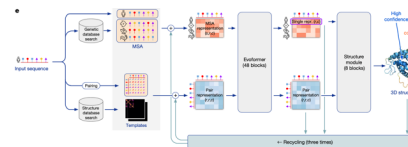
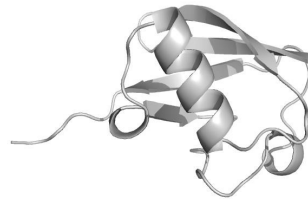
AI used to predict path of particle
(eg. Alonso-Monsalve et al.)

Molecules



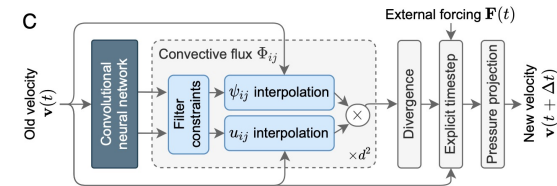
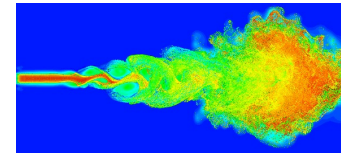
AI used as surrogate model of quantum calculations
(eg. Glick et al.)

Proteins



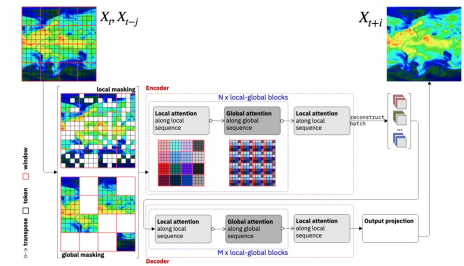
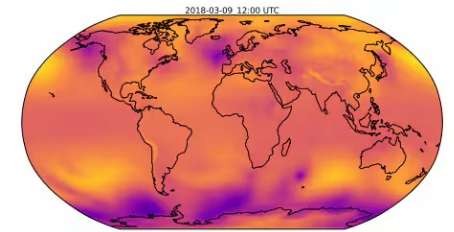
AI used to predict structure from sequence
(eg. Jumper et al.)

Fluid Mechanics



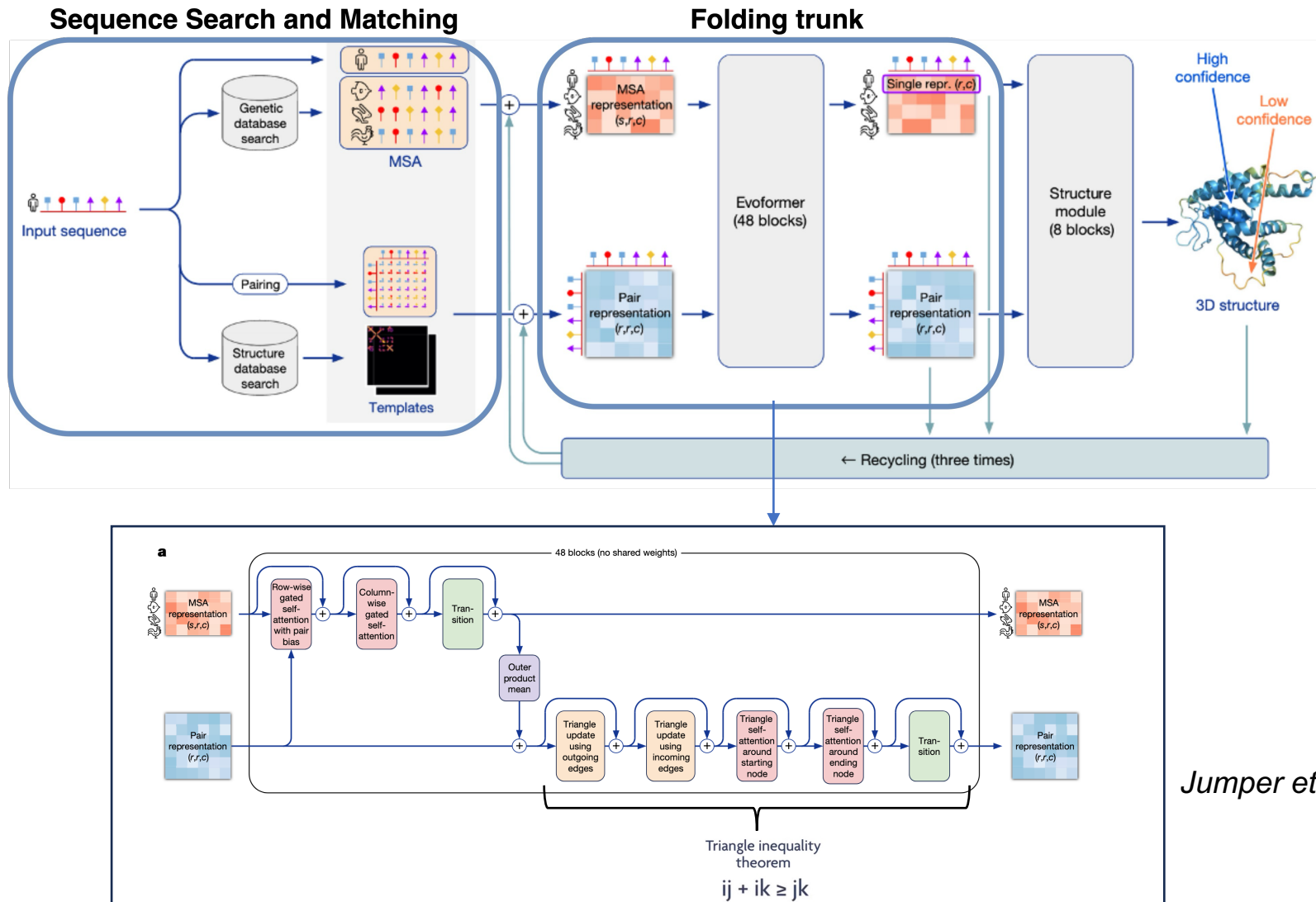
AI used to interpolate from coarse to fine grids
(eg. Kochkov et al.)

Geosciences



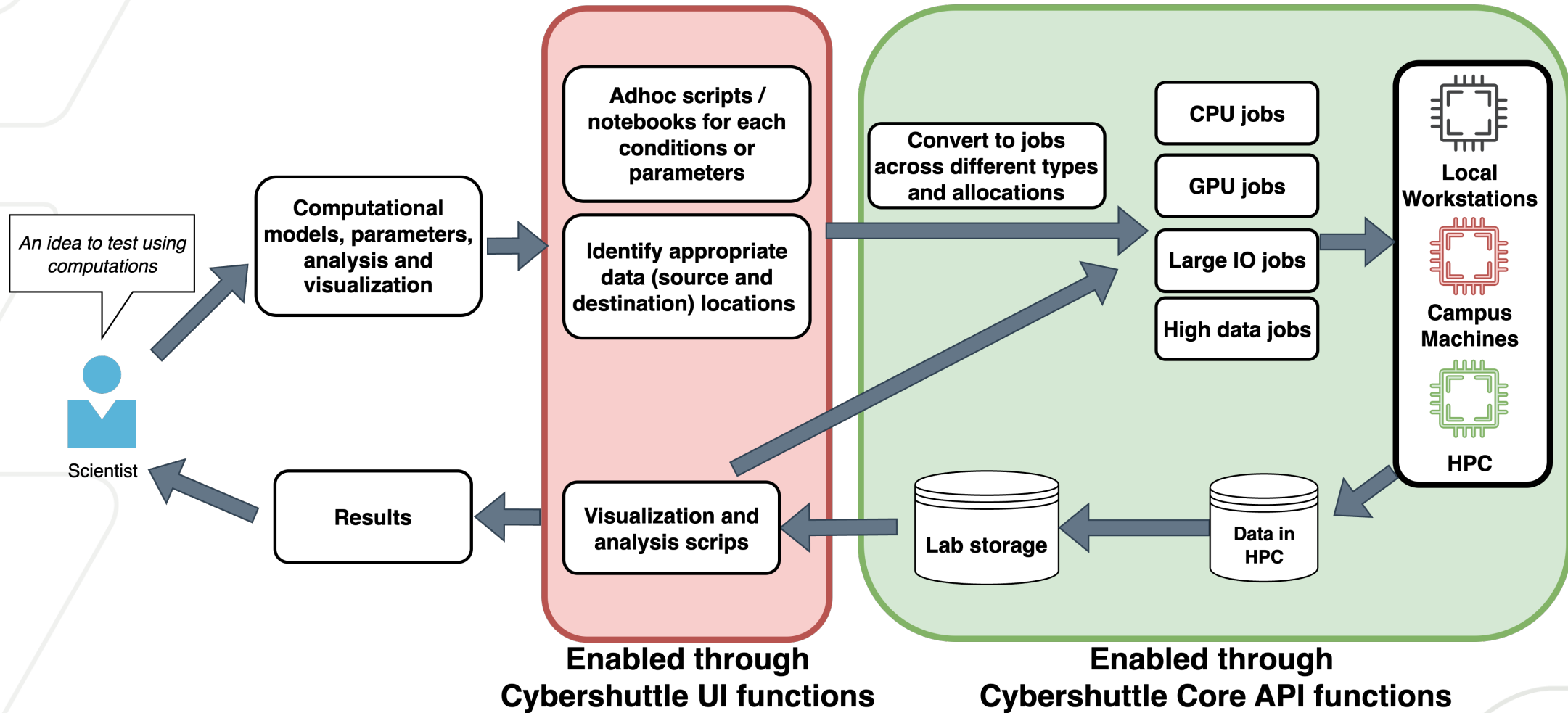
AI used to predict weather
(eg. Schumde et al.)

These are specialized deep neural networks ...



Jumper et al. 2021

Software to enable Multi-scale Computational Science



Bring all this power to research edge devices



Scaling Out

Data and compute sizes beyond what can be processing locally.

Scaling in

Need for Interactivity, Data Privacy, Policies.

Reach out to artisan@groups.gatech.edu